

INDUSTRY PROJECT PROPOSAL

Complete all sections in full. Maximum submission: 5 pages (excluding this cover).

A. PROJECT IDENTIFICATION

A1. Project Title

CocoConnect - A Unified Digital Ecosystem for the Coconut Industry

A2. Project Track (tick one)

☒ UCSC Track (UT) <i>Industry / open-source topic via UGVLE</i>	☐ Industry Track (IT) <i>PDC-registered company project</i>	☐ Startup Track (ST) <i>Student-founded registered startup</i>
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A3. Project Type (tick one)

☒ Software Product <i>Web / <u>mobile</u> / desktop application</i>	☐ Proof of Concept (PoC) <i>Research-based prototype / algorithm</i>
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B. GROUP & SUPERVISORY DETAILS

B1. Student Group

#	Full Name	Registration No.	Degree	Email
1	Diyunugal Pedige Malshi Tharuka Karunathilaka	2023/IS/046	☐ CS ☒ IS	dpmalshitharukakarunathilaka@gmail.com
2	Senuji Nureka Vidanage	2023/IS/108	☐ CS ☒ IS	senujiv1@gmail.com
3	Ibrahim Ifham Kamal	2023/IS/044	☐ CS ☒ IS	ibrahimhejaaz@gmail.com
4	Gigiriwala Gamage Kavithi Senethma Piyamini	2023/IS/074	☐ CS ☒ IS	ggkspiyamini003@gmail.com
5	Kattadige Sachika Pawani Weerawarana	2023/CS/222	☒ CS ☐ IS	psachika4@gmail.com
6	Jameel Fathima Nusrath	2023/IS/070	☐ CS ☒ IS	nusrathjameel2002@gmail.com

i Recommended composition: at least 2 CS students and 2 IS students per group.

B2. Academic Co-Supervisor (UCSC) Mr. Jaliya Amarathunga	B3. Industry Mentor / Company Mr. Shakila Wijewardena
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C. PROJECT OVERVIEW

C1. Problem Statement

i Describe the real-world problem or gap this project addresses. (≈ 100 words)

The coconut industry in Sri Lanka involves a diverse network of growers, processors, buyers, service providers, and estate owners. However, many of its operations still rely on manual processes, phone calls and fragmented systems, resulting in inefficiencies across the agricultural value chain. The major challenges include Lack of price transparency, Inefficient land utilization, Limited access to agricultural services and Fragmented product marketplace.

Existing solutions are siloed, non-digital and fail to serve the broader agricultural ecosystem. Consequently, stakeholders lack a unified platform for collaboration and Ui informed decision-making, while the absence of a transparent and technology-driven environment discourages new investors and entrepreneurs from entering the sector. This highlights the need for an integrated digital platform that connects all stakeholders, improves market accessibility, and promotes sustainable agricultural growth.

C2. Proposed Solution & Objectives

CocoConnect is a mobile-first platform that digitally connects Sri Lanka's coconut industry by bringing together growers, buyers, service providers, estate owners, and harvest recorders on a single platform. It features a product marketplace, a service marketplace, estate management tools, and an admin portal for stakeholder verification and platform management.

Project Objectives

1. Develop a smart marketplace connecting buyers with verified sellers.
2. Create a location-based platform for agricultural service providers.
3. Implement real-time harvest recording and inventory updates.
4. Optimize estate land utilization through digital management tools.
5. Build a scalable and offline-capable ecosystem for the coconut industry.

C3. Key Features / Modules

#	Feature / Module	Brief Description	Primary ILO(s)
1	Stakeholder & User Management	Secure role-based user registration and authentication.	Secure system design and access control.
2	Smart Product Marketplace	Enables buyers to find and purchase products from verified sellers.	Digital marketplace development.
3	Seller & Product Management	Allows sellers to manage products, inventory, pricing, and orders.	Data and business process management.

4	Harvest & Estate Management	Records harvest data and supports estate monitoring with offline access.	Digital resource and operational management.
5	Location-Based Service Marketplace	Connects agricultural workers with nearby service opportunities.	Location-based application development.
6	Order Management & Feedback System	Manages transactions, bookings, ratings, and reviews.	Transaction and quality management.

i List the main features students will individually build. Features/functions of third-party libraries cannot be claimed as contribution.

D. TECHNICAL APPROACH

D1. Technology Stack

Layer	Technology / Framework	Justification
Frontend / UI	Flutter	Cross-platform mobile app development with a single codebase.
Backend / API	Node.js + Express.js	Scalable and efficient REST API development.
Database	PostgreSQL+PostGIS	Reliable data storage with location-based querying support.
DevOps / Deployment	GitHub , Docker , GitHub Actions , AWS/Render	Version control, CI/CD, containerization, and cloud deployment.
Other (e.g. ML, IoT)	JWT, Google Maps API, Notify.lk SMS API	Secure authentication, location services, and SMS OTP verification.

D2. System Architecture Overview

i Provide a high-level architecture diagram or a textual description of how system components interact.

CocoConnect follows a client-server architecture. The Flutter mobile application communicates with the Node.js/Express backend through REST APIs. The backend manages business logic, authentication, and data processing while storing information in a PostgreSQL database with PostGIS for geospatial features. External services such as Google Maps API and Notify.lk SMS API provide location-based functionality and OTP verification, while Docker and GitHub Actions support deployment and CI/CD.

[System Architecture Diagram](#)

E. PROJECT PLAN & WORKLOAD DISTRIBUTION

E1. High-Level Timeline (2 Semesters)

Phase	Milestone / Deliverable	Target Week(s)	Responsible Member(s)
Proposal Milestone	Requirements, scope, architecture, proposal	Weeks 1–2	All Members
Sprint 1	SRS, use cases, database, wireframes	Weeks 3–5	BA & SE Team
Sprint 2	UI/UX design and frontend	Weeks 6–9	SE Team & QA
Sprint 3	Authentication, marketplaces, harvest module	Weeks 10–14	SE Team
Interim Milestone	System integration & presentation	Week 15	All Members
Sprint 4	Estate management & admin portal	Weeks 16–20	SE Team
Sprint 5	Analytics, notifications & offline support	Weeks 21–24	SE Team
Sprint 6	Testing and optimization	Weeks 25–28	SE Team & QA
Final Milestone	Documentation, deployment & final presentation	Weeks 29–30	All Members

i Semester 1 ≈ Weeks 1–15; Semester 2 ≈ Weeks 16–30. Include Proposal, Interim, and Final milestones.

E2. Individual Contribution Plan

Student Name	Primary Responsibilities	Contribution Area	Est. Hours
D.P.M.T.Karunathilaka	Requirements Engineering, Stakeholder Analysis, SRS, Use Case Modelling, Project Documentation	☐ SE ☒ BA ☐ Both	450hrs
S.N.Vidanage	Smart Product Marketplace, Seller & Product Management, Backend API Development	☒ SE ☐ BA ☐ Both	450hrs
G.G.K.S. Piyamini	Service Marketplace, Order Management, Frontend Development	☒ SE ☐ BA ☐ Both	450hrs
K.S.P. Weerawarana	Harvest & Estate Management, Land Capacity Optimizer, Database Integration	☒ SE ☐ BA ☐ Both	450hrs
I.I.Kamal	Authentication & User Management, DevOps, CI/CD Pipeline, Cloud Deployment	☒ SE ☐ BA ☐ Both	450hrs
J. F. Nusrath	Test Planning, System Testing, Integration Testing, Quality Assurance & Validation	☒ SE ☐ BA ☐ Both	400hrs

i SE = Software Engineering | BA = Business Analysis. Each student must contribute ≥ 400 hours across both semesters.

F. INTENDED LEARNING OUTCOME ALIGNMENT

Tick each ILO that is directly addressed by this project and briefly explain how in the space provided.

ILO	Description	Status
ILO-01	Synthesise project requirements into structured execution plans with defined milestones and task distribution.	☒ Addressed
ILO-02	Build functional, scalable software by applying engineering principles, modern frameworks, and deployment strategies.	☒ Addressed
ILO-03	Validate software integrity through rigorous testing and maintain industry-standard technical documentation and version control.	☒ Addressed
ILO-04	Demonstrate professional communication and effective teamwork to achieve project objectives.	☒ Addressed
ILO-05	Resolve complex technical challenges through research and critical thinking while reflecting on professional growth.	☒ Addressed

Brief explanation of ILO coverage:

ILO-01 – Demonstrated through requirements gathering, project planning, SRS preparation, and milestone scheduling.

ILO-02 – Achieved by developing a scalable mobile and web-based system using Flutter, Node.js, and PostgreSQL/PostGIS.

ILO-03 – Demonstrated through testing, technical documentation, GitHub version control, and quality assurance activities.

ILO-04 – Achieved through team collaboration, task allocation, and regular communication with project stakeholders.

ILO-05 – Demonstrated by solving challenges related to location-based services, geospatial data processing, and agricultural ecosystem management.

G. RISKS, CONSTRAINTS & ASSUMPTIONS

#	Risk / Constraint	Likelihood (H/M/L)	Mitigation Strategy
1	Weak GPS signal in rural areas.	H	Use offline caching and sync data when connected.
2	Low digital literacy among growers.	H	Provide local language support and user training. .
3	Payment gateway integration challenges.	M	Use trusted local payment APIs and phased integration.
4	Incorrect harvest data entries.	M	Add approval workflows and data validation checks.
5	Scope expansion beyond project goals.	M	Limit features to Phase 1 and defer advanced functions.
6	Third-party API downtime.	M	Implement fallback services and local data caching.

H. TRACK-SPECIFIC DECLARATIONS

i Complete only the sub-section relevant to your selected track (A2 above). Leave others blank.

H1. UCSC Track (UT)

UGVLE Topic Reference No.: 38	Mentor Name & Organisation: Mr. Shakila Wijewardena Coconut Growers Association of Sri Lanka
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H2. Industry Track (IT)

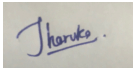
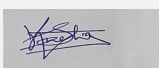
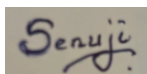
Company Name:	PDC Registration No.:	Agreement Signed? ☐ Yes ☐ No
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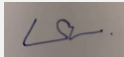
H3. Startup Track (ST)

Registered Company Name:	PDC Registration No.:	Founding Member(s) in Group? ☐ Yes
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I. DECLARATIONS & SIGNATURES

We confirm that the information provided is accurate, that this project is a genuine software development initiative, and that we have read and agree to the Industry Project terms and conditions.

#	Student Name	Signature	Date
1	D.P.M.T.Karunathilaka		05/06/2026
2	I.I.Kamal		05/06/2026
3	J. F. Nusrath		05/06/2026
4	G.G.K.S. Piyamini		05/06/2026
5	K.S.P. Weerawarna		05/06/2026
6	S.N.Vidanage		05/06/2026

Academic Co-Supervisor Name: Mr. Jaliya Amarathunga Signature & Date: 05.06.2026 	Industry Mentor / Supervisor Name: Mr. Shakila Wijewardena Signature & Date:  Mr. Gihan Seneviratne Signature & Date:  05.06.2026
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Submit via UGVLE before the published deadline. Late submissions will not be accepted.